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DESIGN THINKING ORIENTED PROPOSAL : MINDCONNECT

SECJ3553 – ARTIFICIAL INTELLIGENCE

SECTION 02

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1.0 Empathize

During this stage, we are focusing on understanding the user's experiences, emotions, and challenges. Empathize involves observing, communicating and listening to users to find useful and meaningful insights into their needs and the problem they face. The goal is to see the problem from a human perspective rather than a technical one.

With MindConnect, empathy is to make sure an artificial intelligence (AI) - driven mental health platform is capable of responding not only to symptoms but also to emotional realities of users that might be struggling in silence. Recent research indicates that youth and young adults show increasing interest in digital mental health tools, especially when the solution feels safe, accessible, and emotionally supportive (Horstkötter et al., 2025).

1.1 Problem and Pain

Many individuals have challenges and difficulty to get access to quality mental health support. Problem and pain that are commonly found includes:

1. Stigma and fear of judgment from surrounding people can lead to hesitation in seeking professional help.
2. Emotional isolation which users often feel that no one truly listens or understands them.
3. Self-diagnosis based on unreliable sources, particularly through social media.
4. Lack of personalized or continuous support, leaving users feeling unheard or forgotten after initial interactions.

These problems and pains prove the need for a reliable and data-driven platform to communicate, understand and support users indefinitely.

1.2 User and stakeholders

The primary target user for this platform are

1. Individuals that are experiencing stress, anxiety or depression.

2. Students or young adults seeking emotional support but uncertain about or uncomfortable with face-to-face therapy.
3. Professional workers that are currently dealing with burnout or mental fatigue.

Moving on to the secondary users or stakeholders:

1. Mental health professional who can collaborate to channel knowledge with the AI-system for early detection.
2. Educational institution or workplace that promotes mental well-being programs.
3. Families or caregivers who want to support loved ones but may not know how.

1.3 Users' Perspective

If an AI solution were to be developed for mental health support, users would likely have several key concerns and expectations. First and foremost, safety and privacy would be paramount. Users want a secure space where they can express their emotions freely without fear of judgment, data misuse, or exposure. Additionally, there may be skepticism about whether an AI system can genuinely empathize and understand their experiences on a human level. Accessibility is another major consideration, as users would expect the AI to provide support anytime and anywhere. Furthermore, users value personalization, preferring guidance and coping strategies that are tailored to their unique situations rather than generic responses. Last but not least, users seek consistency and continuity. They would prefer an AI companion that offers ongoing care and check-ins rather than just a one-time conversation.

1.4 Goal of Empathize Stage

The objective of the Empathize stage is to gain a deep understanding of the emotional, psychological, and practical needs of individuals seeking mental health support. This stage aims to gather meaningful insights that will guide the design of an AI system capable of delivering empathetic, accessible, and personalized therapeutic experiences. The findings from this phase serve as the foundation for the subsequent Define stage, where the problem statement will be further refined based on the insights obtained.

2.0 Define

The phase will transition from empathize to define phase where the focus will shift from simply observing the problem to clearly articulating the challenge that MindConnect AI is designed to solve the problem identified.

2.1 Point-of-View (POV) Problem Statement

The table below describes the full definition of the user, their critical need, and the underlying reason (insight) for that need based on the problem that has been identified.

Element	Description
User	Individuals experiencing stress, anxiety or early-stage depression, particularly young adults and professional workers
Need	Need a way to access immediate, non-judgemental, and continuous support that offers personalized coping strategies
Insight	Because existing mental health services are too costly, suffer from long wait times and the fear of judgment leads to emotional isolation and silence

2.2 Scope and Boundaries

The following points define the operational and ethical limits of the MindConnect AI solution to ensure its targeted functionality and responsible deployment.

1. **Focus Areas:** MindConnect will initially focus on helping users manage and detect their signs of any mental health problem like Generalized Anxiety Disorder (GAD) and Mild-to-Moderate Depression. Early detection is always a top priority for all users.
2. **Ethical Rule:** The AI will never attempt to give a formal clinical diagnosis or replace a human therapist. It is designed only for support, detection and safe hand-offs.
3. **Accessibility:** The solution must be easily accessible by using a mobile device or web browser, ensuring it is available anytime and anywhere.
4. **Data:** It will train the AI model using verified, privacy-protected psychological datasets.

2.3 AI Success Metrics and Key Performance Indicators (KPIs)

The success of the AI component must be measured with these technical targets:

1. **AI Accuracy (Clinical Safety):** To achieve an accuracy rating of 90% or higher when detecting signs of anxiety and depression.
2. **Triage Speed (Emergency Response):** To ensure the time from high-risk input detection to the display of emergency contact information is under 3 seconds.
3. **User Engagement (Therapy Effectiveness):** To maintain an average user session length of 8 minutes and more, indicating the users are engaging meaningfully with the therapy sessions.

3.0 AI Goals & Solution

The integration of Artificial Intelligence (AI) within *MindConnect* aims to enhance the accessibility and effectiveness of mental health support through data-driven analysis and personalized interaction. The primary objectives of the AI component are as follows:

1. To assist in the preliminary diagnosis of mental health conditions based on users' self-reported symptoms and behavioral indicators through intelligent data analysis.
2. To deliver personalized therapy sessions that adapt to the user's mental state, emotional patterns, and historical interactions to ensure targeted and effective psychological support.
3. To create a safe and judgment-free environment where users can communicate openly about their emotions without the fear of social stigma or bias.

To achieve these objectives, *MindConnect* employs several AI technologies designed to replicate and enhance the processes involved in psychological assessment and therapy.

1. AI-Based Diagnostic System

The system utilizes Natural Language Processing (NLP) and Machine Learning (ML) algorithms to interpret and evaluate user inputs in text or voice form (Stryker et al., 2024). By analyzing linguistic patterns, emotional tone, and contextual information, the AI identifies potential mental health conditions such as anxiety, depression, or stress. The diagnostic model is trained on verified psychological datasets to ensure reliability and accuracy in symptom detection.

2. Personalized Therapy Chatbot

The conversational AI module functions as a virtual therapist capable of providing personalized guidance. It is designed based on psychotherapy methods such as Cognitive Behavioral Therapy (Professional, 2025), enabling it to offer customized coping mechanisms, reflective exercises, and motivational dialogue that align with the user's emotional state and mental progress.

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